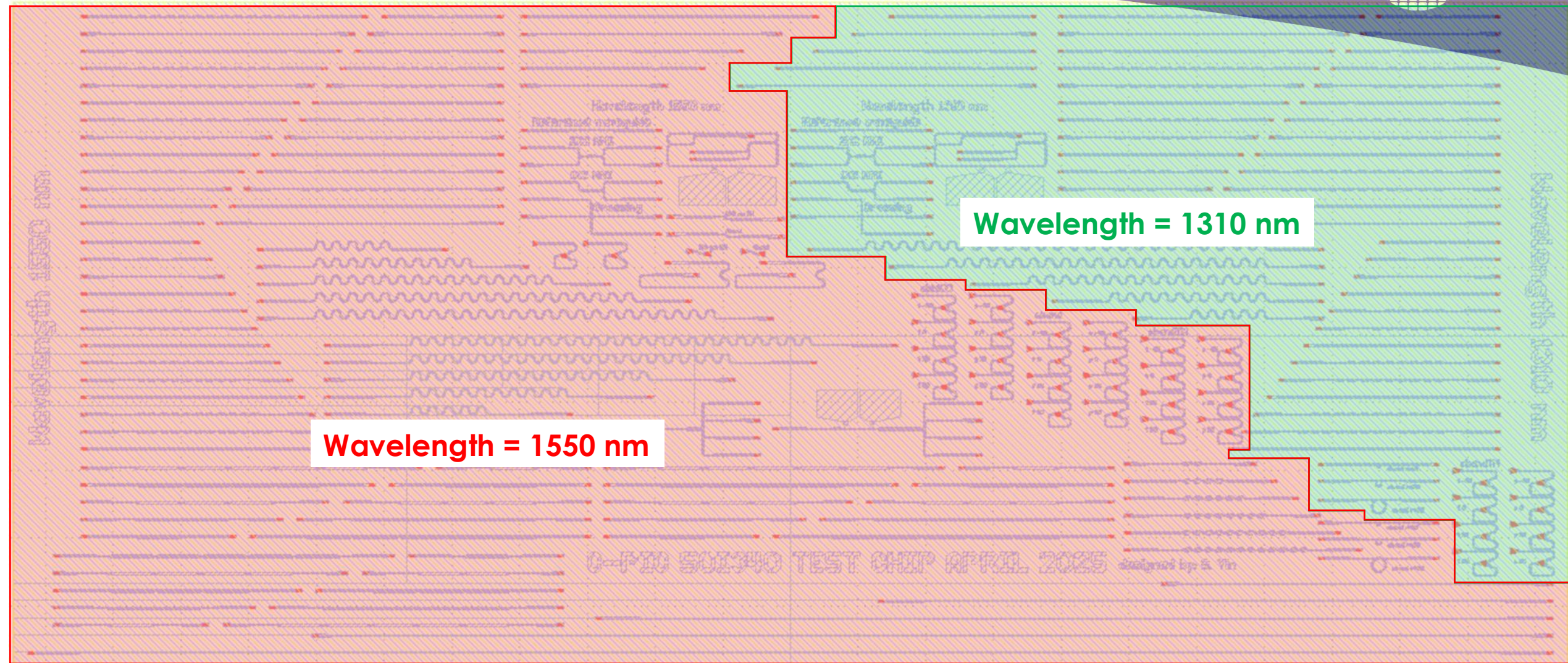




SOI340 TEST CHIP

Designed by Dr Shengqi Yin



No.	Component list (cascading cells)
1	1x2 MMI
2	2x2 MMI
3	1X4 MMI
4	2X4 MMI
5	Crossing
6	Y-splitter
7	90degree-bend
8	Strip-to-rib taper

No.	Component test unit
1	1x2 MMI
2	2x2 MMI
3	1X4 MMI
4	2X4 MMI
5	Crossing
6	Y-splitter
7	Mach-Zehnder interferometer

No.	Parameter sweep group
1	Ring Resonator
2	Apodized grating coupler

2x2 MMI

Crossing

2x2 MMI

1x2 MMI

1x2 MMI

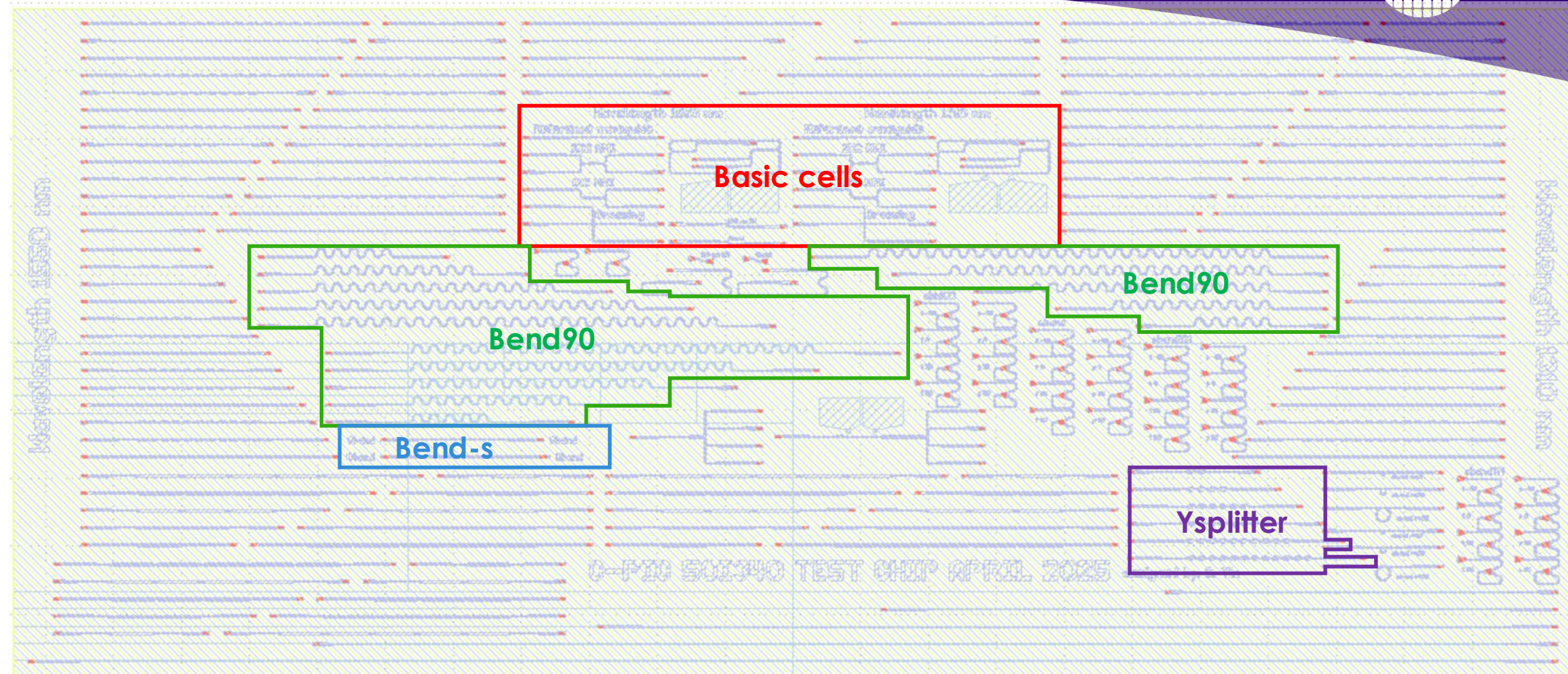
1X4 & 2x4 MMI cell

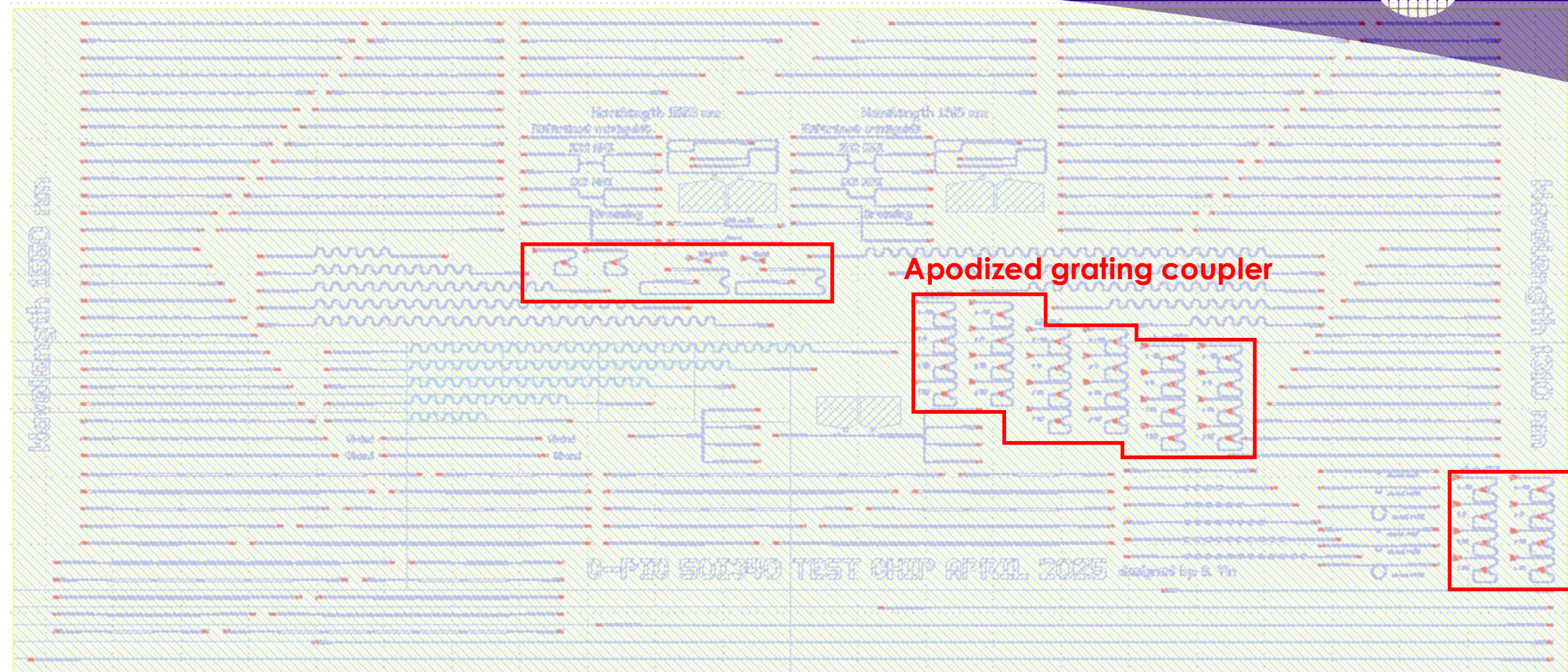
1X4 MMI

2X4 MMI

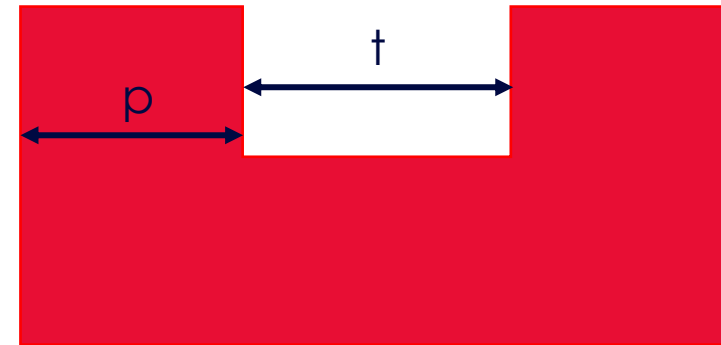
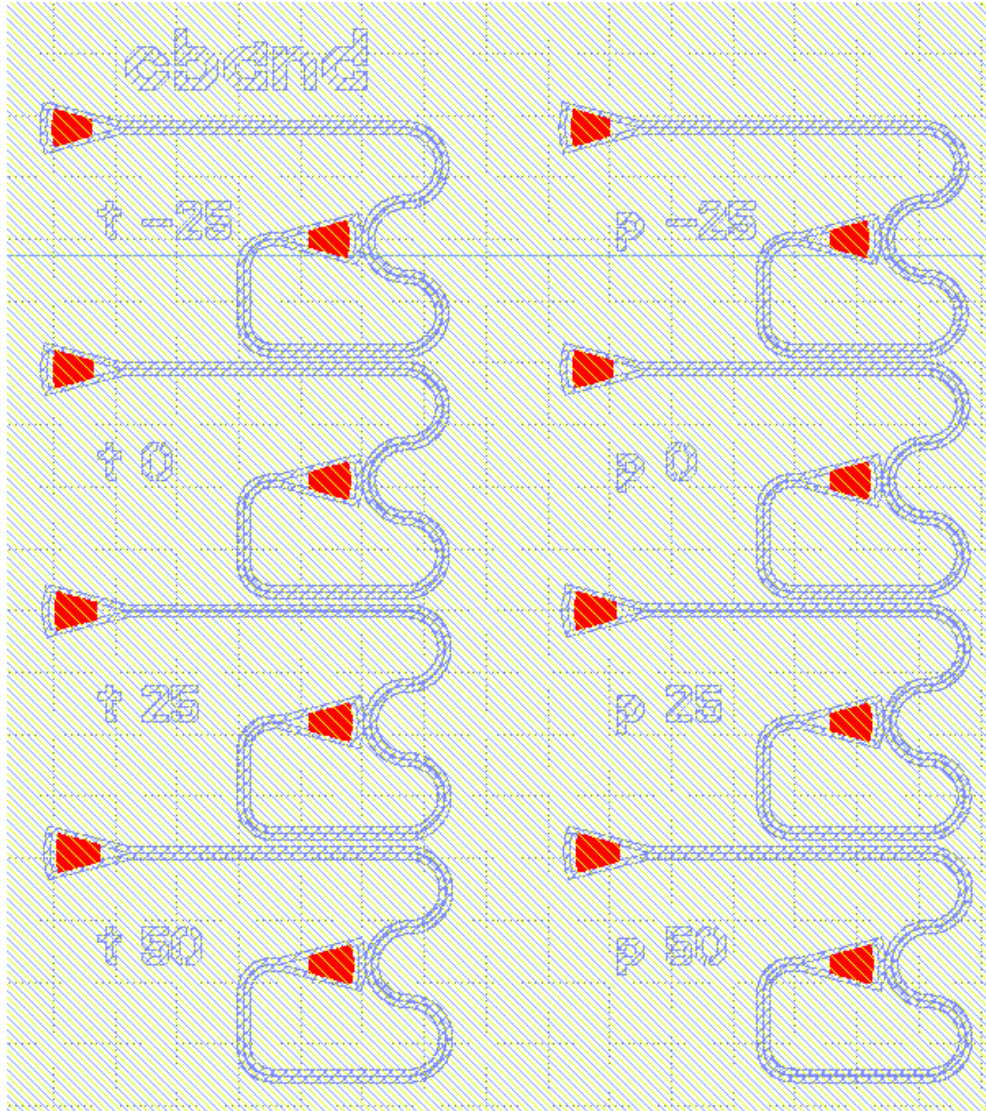
Ring resonator

0-100 501740 TEST CHIP APRIL 2025 designed by S. Yin





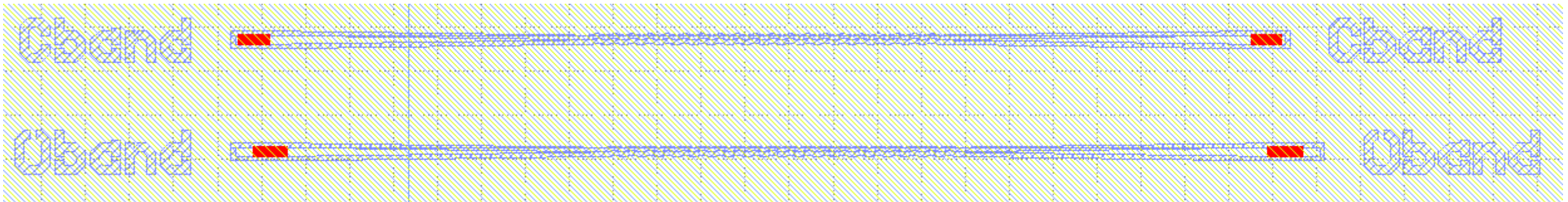
Apodized grating coupler



AGC variation:

- Change etching width (t) and remain width (p)
- Keep minimum resolution (gap and feature) following the DRC rule

Bend-S testing patterns



40 S-bends (used in MMI connections) are cascaded to serve as a reference for propagation loss in the C-band and O-band.